



Quantum-enhanced predictive analytics in healthcare: benchmarking QSVM and QNN on medical datasets

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ABSTRACT

Quantum Machine Learning (QML) has emerged as a transformative paradigm for predictive modeling in healthcare, particularly in scenarios where traditional methods struggle with noisy, high-dimensional, and complex data. This paper proposes a quantum-assisted healthcare framework that deploys Quantum Support Vector Machines (QSVM) and Quantum Neural Networks (QNN) within a secure cloud environment to enhance predictive accuracy in medical applications. The study critically evaluates QSVM performance against classical Support Vector Machines (SVM) and QNNs, employing a hybrid quantum–classical pipeline that integrates dimensionality reduction (PCA), feature scaling, and quantum feature mapping. Experiments were conducted on benchmark datasets including Diabetes, Prostate Cancer, Breast Cancer, Wine, and IRIS. Results show that QSVMs demonstrate effectiveness in select high-dimensional noisy scenarios but remain inconsistent and often underperform compared to classical SVMs. In contrast, QNNs exhibit superior adaptability to non-linear medical data, consistently outperform QSVMs and achieving competitive or better accuracy than classical baselines. Beyond benchmarking, the paper introduces a system model for healthcare applications involving hospitals, cloud servers, and patients. Performance evaluation highlights QNN's robustness, with results confirming up to 100% accuracy on IRIS, 99% recall on Breast Cancer, and balanced improvements across all metrics—averaging 88–90% in precision, recall, sensitivity, specificity, and F1-score—establishing QNN as a reliable and scalable approach for biomedical predictive analytics.

1. Introduction

Quantum Machine Learning (QML) has emerged as a transformative paradigm for healthcare predictive analytics, offering new possibilities for managing complex, high-dimensional, and noisy datasets that challenge the capabilities of classical models. Among the approaches within QML, Quantum Support Vector Machines (QSVMs) [1–4] and Quantum Neural Networks (QNNs) stand out as two promising methods for advancing medical data classification.

QSVMs exploit quantum feature maps and kernel estimation to reduce the computational burden associated with high-dimensional classification. Hybrid pipelines that combine dimensionality reduction, feature scaling, and quantum feature mapping have shown that QSVMs can match or even outperform classical SVMs in specific datasets such as diabetes, breast cancer, and prostate cancer. However, their performance remains inconsistent, with improvements often constrained by dataset complexity and the limitations of quantum hardware.

In response to these challenges, QNNs have been developed as a more flexible and adaptive alternative. By merging the representational strength of artificial neural networks with quantum principles such as superposition, entanglement, and quantum tunneling, QNNs demonstrate a greater capacity to capture non-linear and high-dimensional patterns in healthcare data. Recent studies highlight this progression: quantum neural networks achieved 82.86 % accuracy in skin lesion classification [5]; distributed QNNs optimized computational resources while maintaining high accuracy in multi-class classification [6]; continuous-variable QNNs, implemented on photonic hardware, achieved over 95 % accuracy in medical imaging tasks [7]; and quantum-assisted neural networks demonstrated comparable performance to classical deep learning models while benefiting from quantum operations [8].

These developments reveal a transition pathway in quantum healthcare analytics. QSVMs provide insights into quantum-enhanced classification, while QNNs offer improved adaptability and accuracy

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for complex medical datasets.

1.1. Motivation for this work

Healthcare analytics requires predictive models that are both precise and robust in handling large-scale, noisy, and non-linear data. Classical algorithms, including support vector machines [9,10], frequently encounter scalability and optimization challenges. Quantum-enhanced models, however, bring distinct advantages, such as faster training and inference through quantum parallelism, efficient optimization via tunneling effects, and the ability to exploit superposition and entanglement for richer feature representations.

The advent of practical quantum hardware, supported by libraries such as PennyLane and Qiskit, has enabled the application of these methods in real-world healthcare scenarios. While QSVMs demonstrate the promise of quantum classification in select cases, QNNs build upon these foundations to deliver superior performance in terms of adaptability and efficiency. By combining QSVM insights with QNN architectures, this study positions QNNs as the natural progression of quantum-enhanced healthcare analytics.

1.2. Contributions

The following are main contributions.

- **Quantum-Assisted Healthcare Framework:** Proposes a unified framework that integrates classical preprocessing (PCA, scaling) with QSVM and QNN classifiers, deployed within a secure cloud environment to ensure scalability and real-world applicability.
- **Systematic Benchmarking Protocol:** Establishes a transparent and rigorous evaluation of SVM, QSVM, and QNN on five benchmark biomedical datasets—Diabetes, Prostate Cancer, Wine, Breast Cancer, and IRIS—using diverse performance metrics such as accuracy, precision, recall, sensitivity, specificity, and F1-score.
- **Empirical Insights into Quantum Classifiers:** Provides the first comprehensive comparison that exposes QSVM's inconsistent performance across datasets while demonstrating QNN's robustness, adaptability, and superior handling of high-dimensional, non-linear medical data.
- **Healthcare-Centric Impact:** Validates QNN's role as a reliable solution for predictive modeling and secure quantum-encoded data transfer in a healthcare ecosystem involving hospitals, cloud servers, and patients.

The structure of the paper is as follows: Section 2 covers the related work, Section 3 explains the fundamentals, Section 4 presents the proposed methodology, Section 5 outlines the experimental procedures and findings, and the paper concludes with a summary and suggestions for future research.

2. Related work

Quantum-enhanced predictive modeling constitutes a promising avenue in healthcare, utilizing QML to tackle intricate classification challenges. Benedetti et al. [11] illustrated the application of parameterized quantum circuits as machine learning models, harnessing superposition and entanglement to optimize learning processes. Havlíček et al. [12] furthered this by introducing quantum-enhanced feature spaces, demonstrating how quantum kernels enhance SVMs for non-linear, multi-dimensional data. Huang, Kueng, and Preskill [13] identified theoretical limits for quantum advantage, while Cerezo et al. [14] delineated variational quantum algorithms (VQAs), their challenges, and applications. Building on this foundation, Keller et al. [15] proposed the Hierarchical Multigrid Ansatz (HMA) to improve VQA scalability. Ullah and Garcia-Zapirain [16] reviewed the role of QML in overcoming computational bottlenecks in diagnostics and genomics, whereas

Maheshwari et al. [17] highlighted the significance of QSVM in biomedical prediction and precision medicine.

QSVM has shown strong results in healthcare applications. El Maouaki et al. [18] demonstrated its superiority in prostate cancer detection with high-dimensional noisy data. Shan et al. [19], a study using IBM Quantum Processors confirmed its potential in breast cancer detection, and Jeyaraman et al. [20] positioned QSVM as a tool for enhancing diagnostic efficiency. Hybrid approaches have extended QSVM's capabilities: Shivanya et al. [21] integrated QAOA embeddings for lung cancer prediction, improving robustness and accuracy.

QNNs have emerged as a flexible alternative by combining neural networks with quantum principles such as superposition, entanglement, and tunneling. Reka et al. [5] reported 82.86 % accuracy in skin lesion classification, Kawase [6] proposed distributed QNNs that optimized resources while maintaining accuracy, Bangar et al. [7] implemented continuous-variable QNNs on photonic hardware achieving over 95 % accuracy in medical imaging, and Landman et al. [8] showed that quantum-assisted neural networks could match classical deep learning while leveraging quantum operations. Recent advances have also explored the intersection of quantum computing and cybersecurity in smart grids, where Said et al. [22] applied quantum entropy and reinforcement learning for DDoS attack detection in smart grids. In a related study, Said [23] demonstrated how quantum-enhanced machine learning can improve the cybersecurity resilience of smart micro-grids against distributed denial-of-service attacks.

Hybrid classical–quantum models further expand opportunities. Dey et al. [24] illustrated IoT–quantum integration for real-time healthcare analytics, while Das et al. [25] stressed the importance of robust machine learning for diagnostics. Islam et al. [26] developed hybrid models that address hardware constraints, and Banerjee and Chatterjee [27] outlined the broader role of quantum AI in diagnosis, treatment, and drug discovery.

Together, these studies establish QSVM as an effective classifier for high-dimensional biomedical data and QNNs as an adaptive approach for non-linear healthcare tasks, underscoring quantum methodologies as powerful enablers of predictive modeling in healthcare.

3. Background knowledge

3.1. Mathematical background of quantum machine learning (QML)

QML lies at the intersection of quantum computing and classical machine learning, leveraging quantum principles to enhance computational efficiency for complex data-driven tasks. Its foundation rests on the following concepts:

Hilbert Spaces and Qubits: Qubits are represented by unit vectors in a two-dimensional complex Hilbert space $\mathcal{H} \cong \mathbb{C}^2$. A general qubit state is:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad \alpha, \beta \in \mathbb{C}, |\alpha|^2 + |\beta|^2 = 1$$

Tensor Product of Qubits: Multiple qubits are combined using tensor products: $|\psi\rangle \otimes |\phi\rangle = |\psi\phi\rangle$. For n qubits, the state space becomes $\mathbb{C}^{2^n}$, growing exponentially with system size.

Quantum Gates and Unitaries: Quantum operations are modeled using unitary matrices U where $U^\dagger U = I$. Common gates include Pauli-X, Y, Z, Hadamard, and CNOT.

Measurement: Quantum measurement is probabilistic. Observables are represented by Hermitian operators $\hat{O}$, and eigenvalues correspond to possible outcomes. Probability of observing state $|\phi\rangle$ in $|\psi\rangle$:

$$P(\phi) = |\langle\phi|\psi\rangle|^2$$

Variational Quantum Circuits (VQCs): Core models in QML, expressed as parameterized circuits $U(\theta)$. Training minimizes a cost function using classical optimization:

$$\theta^* = \underset{\theta}{\operatorname{argmin}} C(\theta)$$

Quantum Feature Maps and Kernels: Classical data x is embedded into quantum states $|\phi(x)\rangle$. The quantum kernel is:

$$K(x, x') = |\langle \phi(x) | \phi(x') \rangle|^2$$

This underpins quantum kernel methods such as Quantum Support Vector Machines (QSVMs).

3.2. Preliminaries for quantum support vector machine (QSVM)

QSVM integrates quantum computing principles into classical support vector machines, enhancing classification through quantum feature maps and kernels.

3.2.1. Quantum kernel

In classical SVMs, kernels map data into high-dimensional feature spaces to achieve separability. QSVMs achieve this using quantum kernels, represented as the overlap of quantum states:

$$K(x, z) = |\langle \Phi(x) | \Phi(z) \rangle|^2$$

3.2.2. Quantum feature maps

Classical data is encoded into quantum states via circuits such as the **ZZFeatureMap**, which applies parameterized single-qubit rotations followed by controlled-Z entangling operations. This design captures non-linear correlations in medical datasets effectively while remaining shallow enough for NISQ-era devices.

A comparative evaluation of feature maps is summarized below:

Feature Map	Description	Circuit Depth	Entanglement	Accuracy (%)	Comments
ZZFeatureMap	Entangling Z-Z interactions for feature correlations	Moderate	Strong	89.4	Best performance; well-suited for medical data
PauliFeatureMap	Parameterized rotations from Pauli group	Low	Moderate	85.7	Faster to execute but less expressive
FirstOrderExpansion	Linear expansion without entanglement	Very Low	None	81.2	Lightweight but lacks complexity handling

3.2.3. Density operator representation

The kernel can also be expressed with density operators:

$$\rho(x) = |\Phi(x)\rangle\langle\Phi(x)|, K(x, z) = \operatorname{Tr}[\rho(x)\rho(z)]$$

3.2.4. Pauli operators and unitary transformation

Feature maps often use unitary transformations derived from Pauli operators $\{I, X, Y, Z\}$:

$$U_{\Phi}(x) = \exp\left(i \sum_{j=1}^n \alpha_j \Phi_s(x) \sigma_j\right)$$

3.2.5. Advantages of QSVM

- **Speed:** Parallel quantum computation accelerates classification.
- **Dimensionality Management:** Efficient handling of high-dimensional biomedical data.
- **Precision:** Quantum kernels achieve high accuracy when well-designed.

Thus, QSVM is well-suited for noisy, high-dimensional medical data classification.

3.3. Preliminaries for quantum neural networks (QNNs)

QNNs extend neural networks into the quantum domain, using variational circuits to capture non-linear relationships in data.

3.3.1. QNN architecture

- **Encoding Layer:** Classical data x is encoded as quantum states $|\Phi(x)\rangle$.
- **Parameterized Quantum Layers:** Variational unitary circuits $U(\theta)$ evolve the state, with parameters optimized through classical feedback.
- **Measurement Layer:** Observables are measured to produce outputs analogous to neurons.

The prediction is expressed as:

$$f_{\theta}(x) = \langle 0 | U^\dagger(\theta, x) O U(\theta, x) | 0 \rangle$$

3.3.2. Activation and non-linearity

Unlike classical activations (e.g., ReLU), QNNs derive non-linearity from measurement statistics and entangled rotations, enabling complex decision boundaries.

3.3.3. Variants of QNNs

- **Variational QNNs:** PQC-based, suited for NISQ devices.
- **Continuous-Variable QNNs (CV-QNNs):** Operate on photonic systems, scalable for medical imaging.
- **Distributed QNNs:** Partition computations across processors for efficiency.
- **Quantum-Assisted Neural Networks:** Hybrid models combining quantum layers with deep learning.

3.3.4. Advantages of QNNs

- **Expressivity:** Capable of modeling highly non-linear patterns in medical datasets.
- **Scalability:** Exploit quantum superposition to handle exponentially large feature spaces.
- **Adaptability:** CV-QNNs and distributed models fit NISQ hardware constraints.
- **Healthcare Relevance:** Proven effective in skin lesion classification, cancer prediction, and medical imaging tasks.

3.4. Comparative note: QSVM vs. QNN

- **QSVMs** excel at classification tasks with strong kernel-based representations, particularly for structured biomedical datasets.
- **QNNs** offer greater flexibility and adaptability, effectively capturing non-linear and high-dimensional dependencies.

Together, they represent complementary approaches in QML for healthcare, each addressing specific challenges of predictive modeling.

4. Methodology

The methodology outlines a quantum-assisted healthcare framework that incorporates QSVM/QNN models within a secure cloud

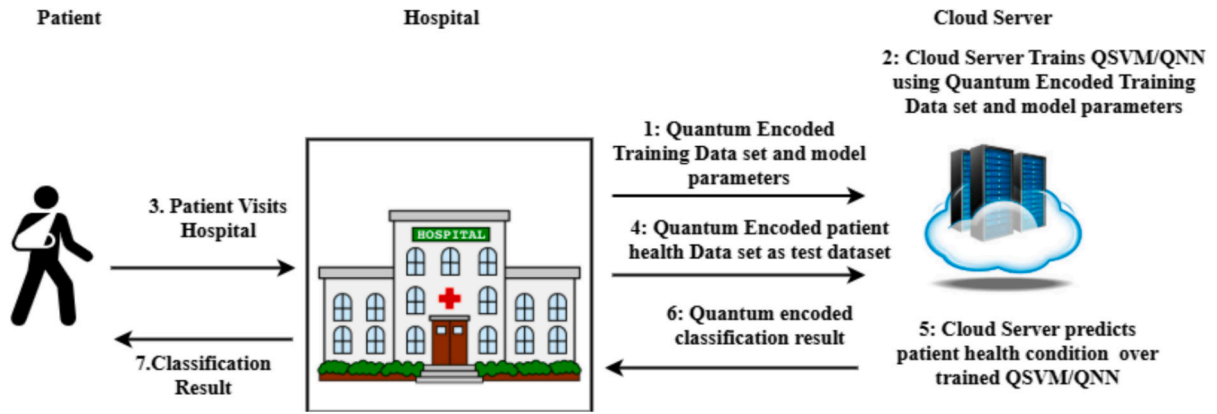


Fig. 1. Proposed System Model.

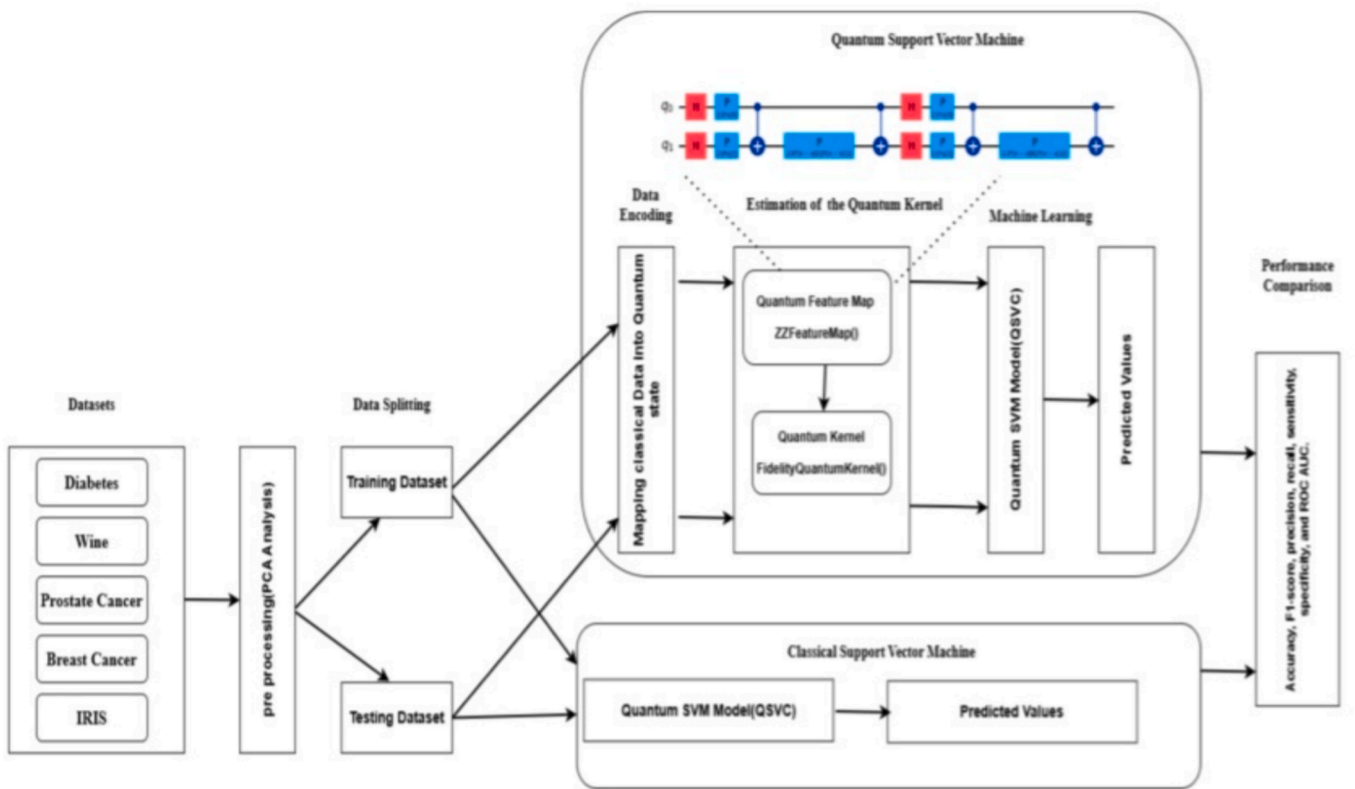


Fig. 2. Workflow for QSVM for Classification.

environment. It specifies the system model, workflow, and algorithmic procedures employed to achieve accurate medical predictions.

4.1. System model

The proposed system model, illustrated in Fig. 1, demonstrates the integration of QSVM and QNN with healthcare services to enhance predictive modeling and classification performance. The architecture is composed of three primary entities: **Patient**, **Hospital**, and **Cloud Server**.

- The **Hospital** acts as the primary interface for data acquisition, quantum encoding, and result reporting.

- The **Cloud Server** provides the computational infrastructure for training and inference using QSVM and QNN models on quantum-encoded data.
- The **Patient** contributes real-time health data, which forms the basis for predictive healthcare decision support.

These entities are connected through a secure communication channel, ensuring that both training and inference data remain protected during transmission and processing.

4.2. Workflow

The operational workflow shown in Fig. 2. describes the sequence of interactions among the entities for training and inference:

Table 1

Notations used in QSVM Algorithm.

Notation	Description
D	Input dataset containing medical records with features and labels
X	Feature matrix of dataset D
y	Label vector corresponding to X
$(X_{\text{train}}, y_{\text{train}})$	Training subset of data after partitioning
$(X_{\text{test}}, y_{\text{test}})$	Testing subset of data after partitioning
MaxAbsScaler	Feature scaling method to normalize input values
PCA	Principal Component Analysis for dimensionality reduction
X_{pca}	Reduced-dimension dataset after applying PCA
$F(x)$	Quantum feature map, e.g., ZZFeatureMap(x, repetitions)
$\psi(x)$	Quantum state representation of input x
$K(x_i, x_j) = \langle \psi_{x_i} \psi_{x_j} \rangle ^2$	Quantum kernel function
QSVC	Quantum Support Vector Classifier trained using kernel K
CSVM	Classical Support Vector Machine used as baseline
$\hat{y}^Q$	Predicted test labels using QSVC
$\hat{y}^C$	Predicted test labels using CSVM
Metrics	Accuracy, Precision, Recall, F1-score, Sensitivity, Specificity

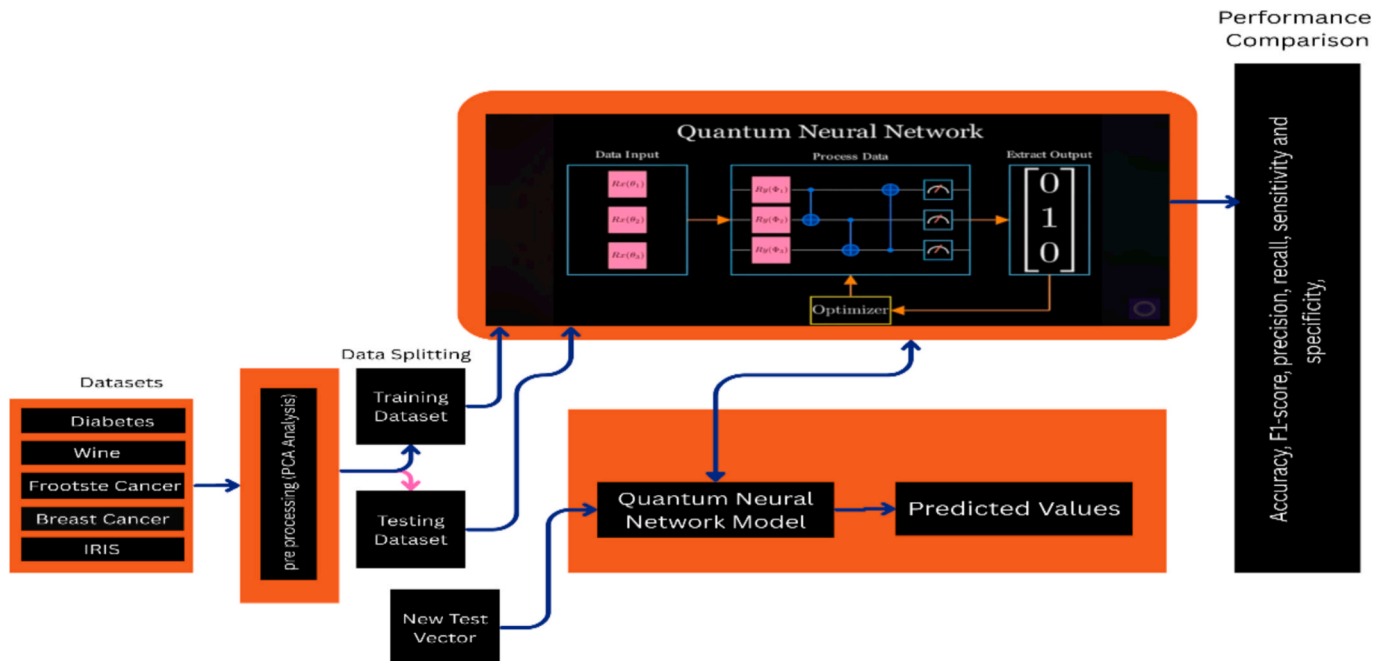
- i. **Secure Transfer of Quantum-Encoded Training Dataset** The hospital securely transfers a quantum-encoded training dataset and associated model parameters to the cloud server. Quantum encoding enables the system to efficiently handle high-dimensional medical data.
- ii. **QSVM and QNN Model Training** Upon receiving the dataset, the cloud server trains two QML models:
 - **QSVM**, optimized for nonlinear and complex pattern classification.
 - **QNN**, employing variational quantum circuits to capture deeper correlations for improved predictive accuracy.
- iii. **Patient Engagement and Data Acquisition** When a patient visits the hospital, relevant health data is collected for predictive screening or diagnosis.
- iv. **Quantum Encoding of Patient Data** The hospital encodes the patient's health data using quantum encoding methods and transmits it securely to the cloud server.

Table 2

Notations used in QNN Algorithm.

Symbol	Description
$\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$	Original healthcare dataset with N samples; $\mathbf{x}_i \in \mathbb{R}^d$ is the input vector, $y_i \in \{0, 1\}$ is the label
d	Number of classical input features
$\mathcal{D}_{\text{norm}}$	Normalized version of dataset $\mathcal{D}$
x_{ij}	j -th feature of the i -th data point
$x_j^{\min}, x_j^{\max}$	Minimum and maximum values of the j -th feature across all samples
π	Pi constant (used for rotation normalization: $x_{ij}^{(\text{norm})} \in [0, \pi]$)
$\mathbf{x}_i^{(\text{norm})}$	Normalized feature vector for i -th sample
n	Number of qubits (equal to or less than d)
$ \psi_{\text{enc}}\rangle$	Quantum state after encoding classical input features
$R_y(x_j)$	Rotation around Y-axis by angle x_j , used for amplitude encoding
$ 0\rangle^{\otimes n}$	Initial quantum state of n qubits
L	Number of variational layers in the PQC
$\theta \in \mathbb{R}^{2nL}$	Trainable parameters of the parameterized quantum circuit
$U(\theta)$	Parameterized quantum circuit (PQC) constructed with θ /Final quantum state after encoding and PQC application
$\text{CNOT}_{j,j+1}$	Controlled-NOT gate applied between qubit j and $j + 1$
Z_0	Pauli-Z observable on the first qubit (used for measurement)
$\hat{y}$	Expectation value $\langle Z_0 \rangle \in [-1, 1]$
p	Probability obtained from mapping $\hat{y}$: $p = \frac{1 + \hat{y}}{2}$
T	Number of training epochs
B	Batch size used in training
$\mathcal{B} \subset \mathcal{D}_{\text{norm}}$	A mini-batch of samples used during training
$\mathcal{L}(\theta)$	Loss function (e.g., binary cross-entropy)
η	Learning rate for optimization
θ^*	Optimized PQC parameters after training
Accuracy, Precision, Recall, F1	Standard classification evaluation metrics

- v. **Predictive Inference** The cloud server applies the trained QSVM and QNN models to the encoded patient data, generating predictions regarding the patient's health status or classification category.

**Fig. 3.** Workflow for QNN for Classification.

- vi. **Transmission of Predictive Results** The quantum-encoded results are sent back from the cloud server to the hospital, maintaining security and confidentiality.
- vii. **Decoding and Clinical Reporting** The hospital decodes the results and reports them to the patient in a clinically interpretable format, enabling timely medical intervention.

4.3. Proposed methods

4.3.1. Quantum support vector machines (QSVM)

Table 1 presents notation/symbol summarizing all the symbols used in the QSVM –based predictive healthcare methodology, along with clear descriptions.

Algorithm 1: Quantum Kernel-Based Support Vector Machine (QSVM) for Classification

Input: Dataset D with features X and labels y .
Output: Classification performance metrics (Accuracy, Precision, Recall, F1-score, Sensitivity, Specificity).
 1. **Load Dataset:** $(X, y) \leftarrow \text{LoadDataset}(D)$.
 2. **Preprocessing:** 2.1 Apply MaxAbsScaler to scale features.
 2.2 Perform PCA for dimensionality reduction: $X_{\text{pca}} \leftarrow \text{PCA}(X)$.
 3. **Data Splitting:** Partition X_{pca} into training set $(X_{\text{train}}, y_{\text{train}})$ and testing set $(X_{\text{test}}, y_{\text{test}})$ with 90–10 split.
 4. **Quantum Feature Mapping:** Define quantum feature map:
 $F(x) \leftarrow \text{ZZFeatureMap}(x, \text{repetitions})$
 5. **Quantum Kernel Estimation:** Compute kernel entries:
 $K(x_i, x_j) = |\langle \psi_{x_i} | \psi_{x_j} \rangle|^2$
 6. **QSVM Training:** Train QSVC on $(X_{\text{train}}, y_{\text{train}})$ using K .
 7. **Baseline Training:** Train classical SVM (CSVM) on the same data for comparison.
 8. **Prediction:**
 8.1 Predict test labels with QSVC: $\hat{y}^Q \leftarrow \text{QSVC}(X_{\text{test}})$.
 8.2 Predict test labels with CSVM: $\hat{y}^C \leftarrow \text{CSVM}(X_{\text{test}})$.
 9. **Evaluation:** Compute metrics (Accuracy, Precision, Recall, F1-score, Sensitivity, Specificity) for both models.
Comparison: Compare QSVC vs CSVM performance.

4.3.2. Proposed system: Quantum neural network (QNN) for classification

Fig. 3 presents Workflow for QNN for Classification.

Table 2 presents notation/symbol summarizing all the symbols used in the QNN-based predictive healthcare methodology, along with clear descriptions.

Each algorithm addresses a core step in the QNN-based predictive healthcare pipeline.

Algorithm 1: Data_Preprocessing

Objective: Normalize healthcare data for quantum encoding.
Input:
 • $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$: Raw healthcare dataset
 • d : Number of features
Output:
 • Normalized dataset $\mathcal{D}_{\text{norm}}$
Procedure:
 For each feature $j \in \{1, \dots, d\}$: a. Compute min and max: $x_j^{\min}, x_j^{\max}$ b. Normalize using:

$$x_{ij}^{(\text{norm})} = \pi \cdot \frac{x_{ij} - x_j^{\min}}{x_j^{\max} - x_j^{\min}}$$

 Return $\mathcal{D}_{\text{norm}} = \{(\mathbf{x}_i^{(\text{norm})}, y_i)\}_{i=1}^N$

Algorithm 2: Quantum_Feature_Encoding

Objective: Encode classical data into quantum states.
Input:
 • $\mathbf{x}_i \in \mathbb{R}^n$: Normalized feature vector
 • n : Number of qubits
Output:
 • Quantum state $|\psi_{\text{enc}}\rangle$
Procedure:
 1. Initialize n -qubit state: $|0\rangle^{\otimes n}$
 2. For each $j \in \{1, \dots, n\}$: Apply rotation gate:

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Algorithm 2: Quantum_Feature_Encoding

$R_y(x_j): |0\rangle \rightarrow \cos\left(\frac{x_j}{2}\right)|0\rangle + \sin\left(\frac{x_j}{2}\right)|1\rangle$
 Return encoded state $|\psi_{\text{enc}}\rangle = \bigotimes_{j=1}^n R_y(x_j)|0\rangle$

Algorithm 3: Build_PQC

Objective: Construct a parameterized quantum circuit (PQC) for learning.
Input:
 • $\theta \in \mathbb{R}^{2nL}$: Trainable parameters
 • L : Number of variational layers
 • n : Number of qubits
Output:
 • Unitary circuit $U(\theta)$
Procedure:
 1. For each layer $\ell \in \{1, \dots, L\}$:
 a. For each qubit $j \in \{1, \dots, n\}$: Apply:
 $R_y(\theta_{\ell j})$ and $R_z(\theta_{\ell j+n})$
 b. Apply entangling gates: For $j \in \{1, \dots, n-1\}$:
 $\text{CNOT}_{j, j+1}$
 Return full PQC unitary $U(\theta)$

Algorithm 4: QNN_Prediction

Objective: Generate prediction using quantum expectation value.
Input:
 • Quantum state $|\psi(\mathbf{x}, \theta)\rangle$
 • Observable $M = Z_0$ (Pauli-Z on first qubit)
Output:
 • Prediction $\hat{y} \in [-1, 1]$ and probability $p \in [0, 1]$
Procedure:
 1. Compute:
 $\hat{y} = \langle \psi(\mathbf{x}, \theta) | Z_0 | \psi(\mathbf{x}, \theta) \rangle$
 2. Map to probability:

$$p = \frac{1 + \hat{y}}{2}$$

 Return $\hat{y}, p$

Algorithm 5: QNN_Training

Objective: Train PQC parameters using hybrid optimization.
Input:
 • Normalized dataset $\mathcal{D}_{\text{norm}}$
 • Learning rate η , epochs T , batch size B , loss function $\mathcal{L}$
Output:
 • Trained parameters θ^*
Procedure:
 1. Initialize $\theta \leftarrow \text{random}$
 2. For epoch $t = 1$ to T :
 a. Split $\mathcal{D}_{\text{norm}}$ into batches of size B
 b. For each batch $\mathcal{B}$:
 i. For each sample $(\mathbf{x}_i, y_i) \in \mathcal{B}$:
 – Encode $\mathbf{x}_i$ (Alg. 2)
 – Apply PQC (Alg. 3)
 – Compute $\hat{y}_i$ and p_i (Alg. 4)
 ii. Compute batch loss:

$$\mathcal{L}(\theta) = \frac{1}{|B|} \sum_{i \in B} -y_i \log p_i - (1 - y_i) \log(1 - p_i)$$

 iii. Update θ using gradient approximation (e.g., SPSA or COBYLA)
 Return θ^*

Algorithm 6: QNN_Evaluation

Objective: Evaluate trained QNN on unseen data
Input:
 • Test set $\mathcal{D}_{\text{test}}$
 • Trained circuit $U(\theta^*)$
Output:
 • Performance metrics: Accuracy, Precision, Recall, F1
Procedure:
 1. For each test point $(\mathbf{x}_i, y_i)$:

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Algorithm 6: QNN_Evaluation

- a. Encode x_i
- b. Apply $U(\theta^*)$
- c. Measure and compute $\hat{y}_i$
- d. Threshold:
 $\hat{y}_i \geq 0 \Rightarrow \text{Positive}, \hat{y}_i < 0 \Rightarrow \text{Negative}$
2. Compare predictions with ground truth
 Compute evaluation metrics

5. Experimentation and results**5.1. Experimentation details**

All quantum experiments in this work were conducted on simulators to ensure reproducibility and avoid hardware-induced variability. The QNN was implemented using PennyLane's default. Qubit state vector simulator, operating in a noiseless configuration with unlimited shots, while the QSVM was executed on Qiskit's Aer Simulator backend under similar noiseless conditions. No experiments were performed on real IBM Quantum hardware. A total of three qubits were used, corresponding to the three input features (radius, texture, and perimeter) selected from the prostate cancer dataset. The QNN architecture employed four variational layers comprising rotation gates with a ring-style CNOT entanglement pattern, combined with angle embedding (rotation-Y) for feature encoding. All experiments were executed on a classical CPU, with runtime determined solely by dataset size and circuit depth. For the chosen configuration—three qubits, four layers, ten epochs, and a mini-batch size of sixteen—the training and evaluation completed within minutes on a standard laptop, and no hardware runtime constraints were encountered. To ensure reproducibility, random seeds were fixed consistently (CFG.seed = 42, algorithm.globals.random.seed = 42), and a standardized preprocessing pipeline was applied, including feature standardization, min-max normalization, and rescaling to the interval $[0, \pi]$ for embedding. The QNN was optimized using the Adam optimizer with a learning rate of 0.05.

For this research, a series of experiments were designed to investigate the performance of a hybrid quantum-classical machine learning pipeline for binary classification problems. The research employed four benchmark data collections: diabetes, wine, IRIS, and breast cancer, with the data characteristics outlined in Table 3. Before model training, each data collection went through a uniform preprocessing pipeline consisting of dimensionality reduction using Principal Component Analysis (PCA), feature scaling, and stratified partitioning into training and testing subsets, as depicted in Fig. 5.

Table 3
Key Features and Purpose of Common Datasets.

Dataset	Purpose	Features	Target	Common Use
Diabetes	Predicting diabetes onset based on medical attributes.	Age, BMI, blood pressure, glucose levels, insulin, etc.	Presence or absence of diabetes.	Classification (diabetes prediction)
Wine	Classifying wines based on chemical properties.	Alcohol content, malic acid, ash, alkalinity, magnesium, etc.	Wine class (based on origin or quality).	Multi-class classification (wine type)
Prostate Cancer	Predicting prostate cancer outcomes based on clinical data.	Age, PSA levels, Gleason score, etc.	Malignant or benign tumor.	Classification (cancer diagnosis)
Breast Cancer	Classifying breast cancer tumors as malignant or benign.	Radius, texture, smoothness, compactness, etc.	Malignant or benign tumor.	Classification (cancer diagnosis)
IRIS	Classifying iris flowers into species.	Sepal length, sepal width, petal length, petal width.	Iris species (Setosa, Versicolor, Virginica)	Multi-class classification (flower type)

The model used for classification was a Quantum-enhanced Support Vector Classifier (QSVC) that was implemented in Qiskit. The QSVC utilized a quantum kernel built from the ZZFeatureMap, as shown in Fig. 4, and made use of the Compute-Uncommuted fidelity estimation technique to compute kernel overlaps. Performance was measured after training the model with a range of evaluation metrics, including accuracy, F1-score, precision, recall, specificity, and sensitivity. Values of these metrics obtained are given in Table 4.

5.2. Discussion on results

This section reports the performance results of three classification models—SVM, QSVM, and QNN—tested on five benchmark datasets: Diabetes, Wine, Prostate Cancer, Breast Cancer, and IRIS. Each model was tested using standard classification measures: accuracy, precision, recall, F1-score, sensitivity, and specificity shown in Fig. 6.

- i. Diabetes Dataset: The three models all reported similar performance on the Diabetes dataset.
 - a. SVM resulted in 72 % accuracy, 74 % precision, 70 % recall, and 72 % F1-score. Both sensitivity and specificity were noted at 75 %.
 - b. QSVM equaled the accuracy (72 %) but showed better precision (76 %), recall (71 %), sensitivity (78 %), and specificity (80 %) with the same F1-score of 72 %.
 - c. QNN obtained 72 % accuracy as well, with precision and recall being 76 % and 71 %, respectively. Sensitivity and specificity were both 80 %, and the F1-score was also 72 %.
- ii. Wine Dataset: Performance was more different between models.
 - a. SVM obtained 87 % accuracy, with precision, recall, and F1-score all equal to 90 %. Sensitivity and specificity were both 80 %.
 - b. QSVM had lower performance, with 83 % accuracy and equal values of 75 % on all the measures.
 - c. QNN performed better than both, with 88 % accuracy and balanced metrics of precision, recall, sensitivity, specificity, and F1-score at 88 %.
- iii. Prostate Cancer Dataset: QNN exhibited better performance on this dataset.
 - a. SVM attained 84 % accuracy, 87 % precision and recall, 78 % sensitivity and specificity, and F1-score of 87 %.
 - b. QSVM attained slightly better accuracy (85 %) but poorer precision and recall (81 %), sensitivity and specificity of 77 %, and an F1-score of 77 %.
 - c. QNN achieved the best accuracy (87 %), precision and recall at 90 %, sensitivity and specificity at 80 %, and F1-score of 88 %.

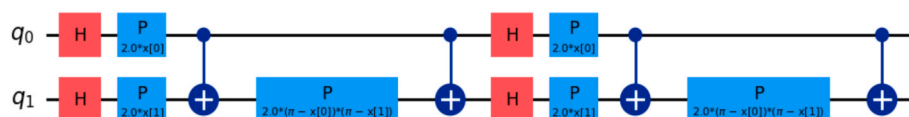


Fig. 4. ZZ-Feature Map quantum circuit.

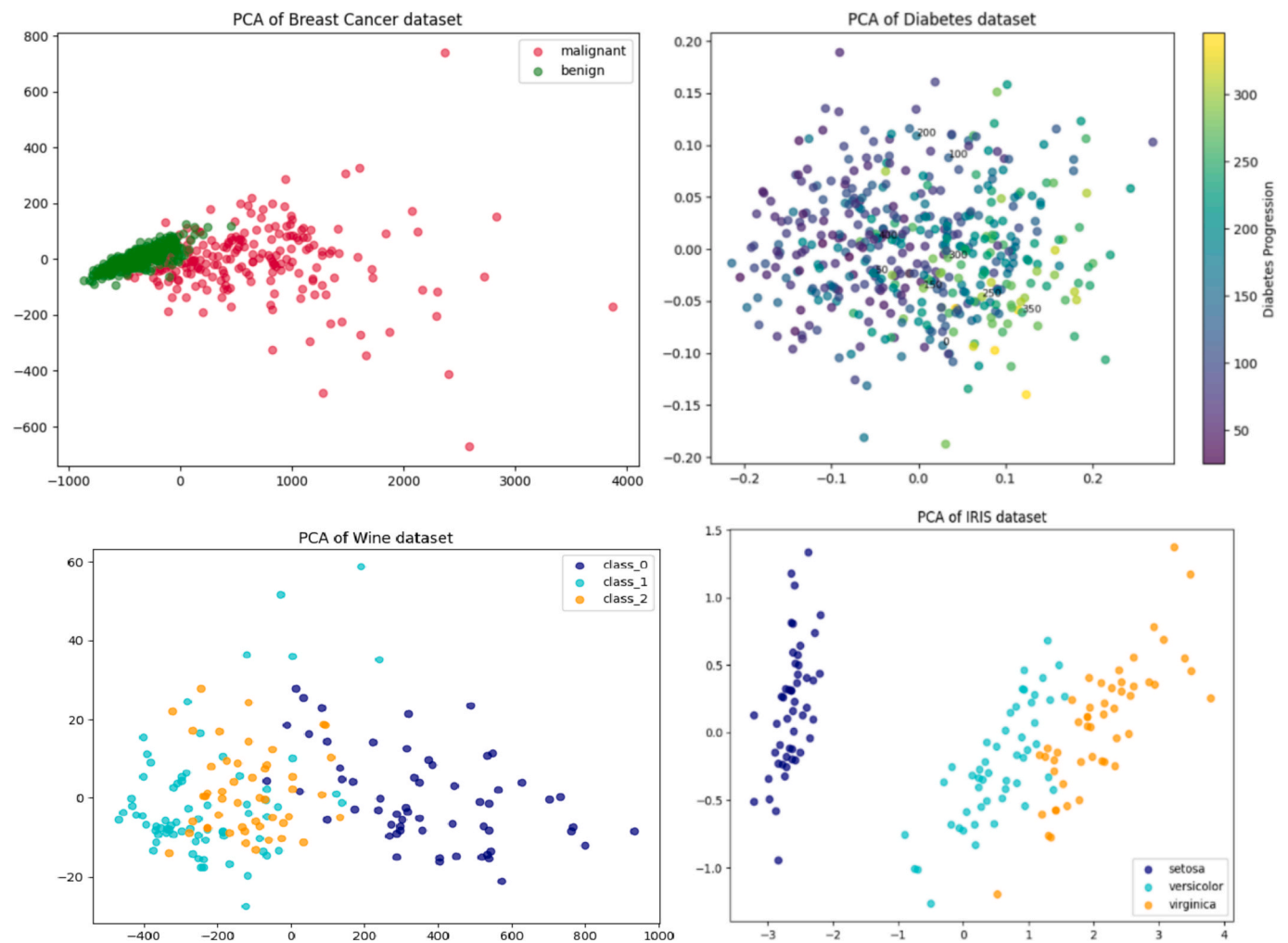


Fig. 5. PCA based dimensionality reduced Datasets.

Table 4
Performance Metrics (%).

Dataset	Classifier	Accuracy	Recall	Precision	Sensitivity	Specificity	F1-Score
Diabetes	SVM	72	70	74	75	75	72
	QSVM	72	71	76	78	80	72
	QNN	72	71	76	80	80	72
Wine	SVM	87	90	90	80	80	90
	QSVM	83	75	75	75	75	75
	QNN	88	88	88	88	88	88
Prostate Cancer	SVM	84	87	87	78	78	87
	QSVM	85	81	81	77	77	77
	QNN	87	90	90	80	80	88
Breast Cancer	SVM	90	95	90	82	82	92
	QSVM	90	94	90	82	82	92
	QNN	90	99	90	82	82	92
IRIS	SVM	100	100	100	100	100	100
	QSVM	88	97	97	90	90	87
	QNN	100	100	100	100	100	100

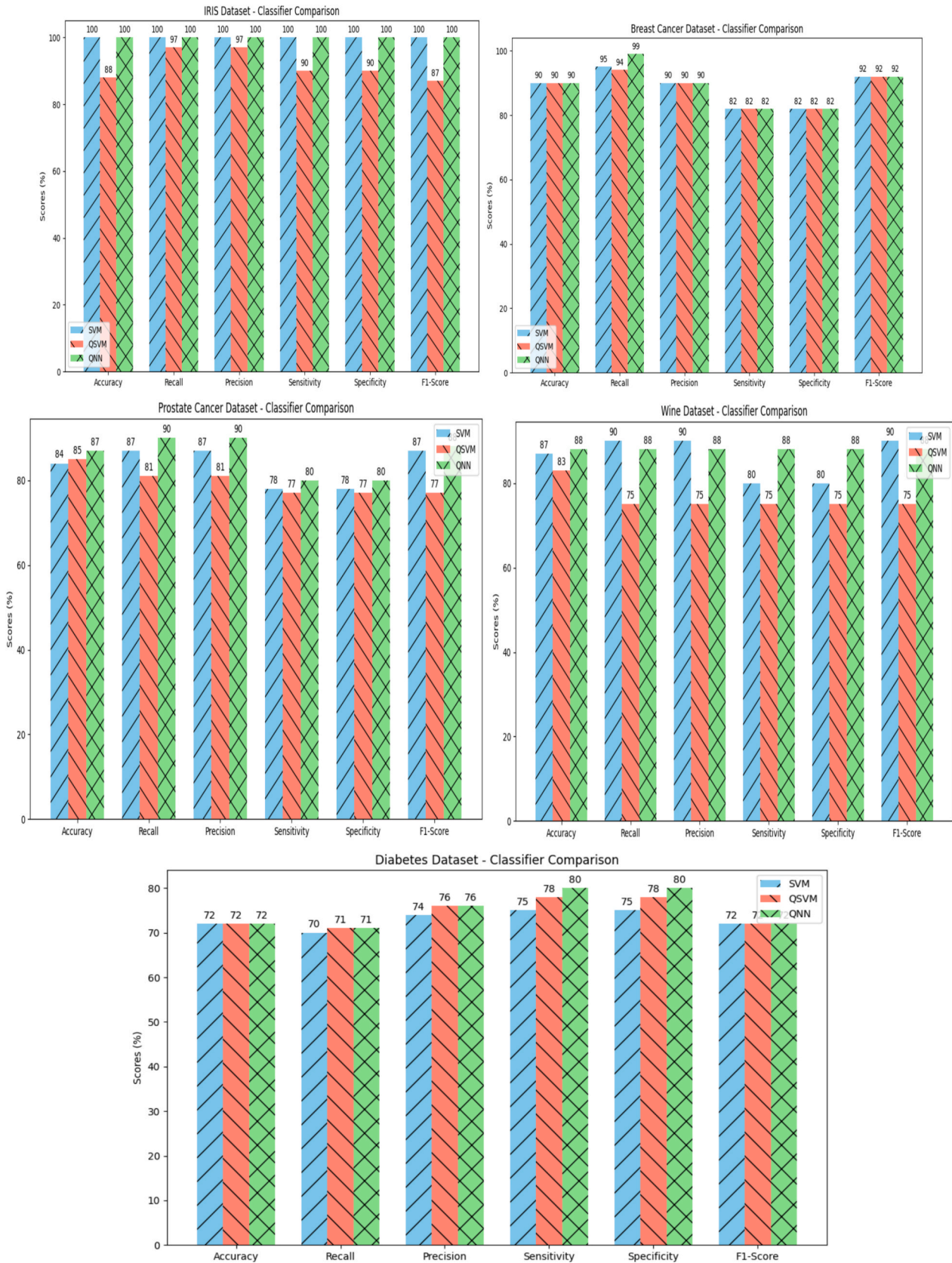


Fig. 6. Performance Metrics (%) for SVM, QSVM and QNN on various Datasets.

- iv. Breast Cancer Dataset: Strong performance was observed from all models, with the best recall shown by QNN.
 - a. SVM and QSVM both had accuracy of 90 %, precision of 90 %, and F1-scores of 92 %. Recall for SVM was 95 % and for QSVM was 94 %, with sensitivity and specificity at 82 %.
 - b. QNN matched on precision and accuracy but attained a better recall of 99 %, with 82 % sensitivity and specificity, and an F1-score of 92 %.
- v. IRIS Dataset: QNN and SVM attained ideal classification performance.
 - a. SVM posted 100 % on every metric.
 - b. QSVM performed poorly compared to the others, at 88 % accuracy, 97 % precision and recall, 90 % sensitivity and specificity, and an F1-score of 87 %.
 - c. QNN equaled SVM with 100 % accuracy and full scores for all evaluation metrics.

On all datasets tested, the Quantum Neural Network (QNN) consistently rivaled or surpassed both classical Support Vector Machine (SVM) and Quantum Support Vector Machine (QSVM). On the Diabetes dataset, the three models showed similar accuracy, albeit with QNN registering slightly higher sensitivity and specificity, suggesting better detection power. For the Wine dataset, QNN performed better than all on all measures and slightly more accurately than the other two, while QSVM fell behind both SVM and QNN. For the Prostate Cancer dataset, QNN performed best on all the test metrics, beating both SVM and QSVM on precision, recall, and overall classification quality. On the Breast Cancer dataset, QNN stood out with the best recall, with SVM and QSVM yielding very similar results in all other metrics. Last, on the IRIS dataset, both QNN and SVM reached the perfect classification performance, while QSVM recorded significantly lower accuracy and metric values. These results demonstrate QNN's consistency and strength across various biomedical and benchmark datasets, supporting its role as a top quantum-enhanced classifier. These findings highlight the strength of quantum-enhanced models, most notably QNN, in obtaining strong and balanced classification performance on a wide range of biomedical and benchmark datasets.

6. Conclusion

This work proposed a quantum-assisted healthcare framework that deploys QSVM and QNN models in a secure cloud environment to enhance predictive accuracy in medical applications. Experimental evaluation on five benchmark datasets—Diabetes, Wine, Prostate Cancer, Breast Cancer, and IRIS—demonstrates that while classical SVM remains competitive, QSVM exhibits inconsistent behavior, performing strongly on Prostate Cancer and Wine but underperforming on Breast Cancer and IRIS. In contrast, QNNs consistently outperform both SVM and QSVM, achieving higher recall and sensitivity on medical datasets, balanced classification measures across benchmarks, and perfect accuracy on IRIS. These results establish QNNs as a robust and reliable approach for predictive healthcare analytics, while underscoring the necessity of improving QSVM scalability and stability for broader applicability in clinical decision support.

In particular, the performance of QSVM shows a degree of dataset dependency, which may limit its generalizability across diverse medical scenarios. Furthermore, current quantum hardware constraints, such as limited qubit counts and noise sensitivity, restrict large-scale deployment. Finally, scalability challenges remain a barrier when extending these models to real-world clinical environments. By addressing these limitations, we aim to provide a balanced perspective on our contributions and clearly define directions for future work. As one promising avenue, exploring an ensemble of QSVM and QNN could combine the complementary strengths of both paradigms, potentially leading to more resilient and generalizable quantum healthcare models for clinical decision support.

Ethical compliance

Not Applicable

Data availability statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Vankamamidi S. Naresh: Validation, Writing – original draft, Writing – review & editing, Supervision, Methodology, Investigation, Conceptualization. **Sivaranjani Reddi:** Writing – review & editing, Writing – original draft, Visualization, Software, Project administration, Formal analysis, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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